Team Control Number

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Problem Chosen

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Summary Sheet

Research on Comprehensive Environmental Assessment of Sport Mega-Events

As public awareness of sustainability grows, **mega sporting events** are facing stricter scrutiny and public pressure. Beyond excitement and huge economic gains, comprehensive evaluations for such events are in urgent need in order to strive for **greener, more responsible decision making**. This is especially obvious when it comes to the case study of **Super Bowl LIX** in New Orleans. Even though the NFL had taken measures to mitigate the environmental footprint, Super Bowl LIX still left a significant impact on the local environment.

In this study, we primarily developed two sets of comprehensive evaluation metrics for the Super Bowl and the World Cup, respectively.

Covering **more than 14 contributing factors**, our **City Impact Evaluation Model (CIEM)**, which evaluates potential host cities for **Super Bowl**, is established based on 5 core indices of environmental impact: Carbon Emission Index (CEI), Energy Consumption Index (ECI), Water Consumption Index (WCI), Air Quality Index (AQI), and Landfill Index (LFI). Next, we determined the weights of each index during multi-criteria decision-making (MCDM) by strategically combining the Analytical Hierarchy Process (AHP) and the Entropy Weight Method (EWM). The AHP-EWM MCDM provides an adaptable, transparent, and structured framework for weighting, ensuring objectivity within subjectivity while yielding a set of balanced, practical results. Finally, we utilized Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) to generate a comprehensive scoring for each potential host, recommending **Inglewood, California**, for future Super Bowl events, which has the highest scoring.

To adapt CIEM to assess the eligibility of potential host countries for the **World Cup**, we established our **National Impact Evaluation Model (NIEM)**. NIEM encompasses 6 core indices of environmental impact: CEI (adapted), ECI (adapted), WCI (adapted), AQI (adapted), LFI (adapted), and a new Environmental-Industrial Index (EII). Similarly, NIEM employs a smart combination of both AHP and Criteria Importance Through Intercriteria Correlation (CRITIC) due to the **interdependence** between the 6 indices; the AHP-CRITIC MCDM guarantees transparency and objectivity within subjectivity and effectively handles the interconnections of the data. Ultimately, TOPSIS is again used to yield a composite score for each potential host country, selecting **Germany** with the highest score as the best choice for hosting future World Cup events.

Notably, we conducted **sensitivity analyses** for CIEM by adjusting the **AHP-to-EWM ratios** and the numbers of participants in a Super Bowl event. First, AHP-to-EWM ratios determine which weighting method is more influential. By slightly fluctuating the ratio, our final scoring for the potential host cities remained approximately unchanged, proving CIEM to be robust and reliable. Second, the **number of participants** is also altered, resulting again in slight fluctuation in the results, further corroborating CIEM's **robustness and reliability**.

Assessing CIEM and NIEM, both integrate a relatively **wide range of contributing factors** that constitute 5 and 6 indices, respectively, ensuring comprehensiveness. Moreover, both models are **adaptable** to a variety of host cities or countries. CIEM can be applied to most US states, while NIEM can be used to evaluate most countries across the world. Nonetheless, due to the lack of data, CIEM and NIEM are both approximation models based on assumptions. If enhanced with data currently unavailable, CIEM and NIEM will be considerably more accurate and practical.

Keywords: Environment, sustainability, Super Bowl, the World Cup, mega-sporting event, evaluation

Contents

1 Introduction **3**

1.1 Background . . . 3

1.2 Problem Restatement . . . 3

2 Assumptions with Justification **4**

3 Environmental Evaluation of Super Bowl LIX **5**

3.1 Establishment of an Environmental Footprint Framework . . . 5

3.2 Quantitative Impact Analysis by Category . . . 5

4 City Impact Evaluation Model (CIEM): Indices **6**

4.1 Carbon Emission Index (CEI) . . . 6

4.2 Energy Consumption Index (ECI) . . . 7

4.3 Water Consumption Index (WCI) . . . 8

4.4 Air Quality Index (AQI) . . . 9

4.5 Landfill Index (LFI) . . . 10

4.6 Computed City Indices Values . . . 11

5 City Impact Evaluation Model (CIEM): Weighting and Evaluation **11**

5.1 Subjective Weighting: Analytic Hierarchy Process (AHP) . . . 11

5.2 Objective Weighting: Entropy Weight Method (EWM) . . . 12

5.3 Comprehensive Evaluation: Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) . . . 14

5.4 Results . . . 14

6 Sensitivity Analysis **14**

6.1 Weight parameter α sensitivity analysis . . . 14

6.2 Number of Participants Sensitivity Analysis . . . 16

7 National Impact Evaluation Model (NIEM) **17**

7.1 Addition and Adjustments of Indices . . . 18

7.2 Weighting . . . 20

7.3 Comprehensive Evaluation (TOPSIS) and Results . . . 22

7.4 Comparison and Conclusion . . . 22

8 NFL Host City Letter **23**

9 Strengths and Weaknesses **24**

10 AI Use Demonstration **26**

1 Introduction

1.1 Background

As technology develops, an increasing amount of population is paying attention to the environmental footprint and sustainability behind major sporting events, trying to make the world a better place for all life. Major sporting events such as the Super Bowl and the World Cup generate great economic revenue, but along with great revenue, there are great sacrifices to the environment due to logistics, waste and energy behind the scenes. Although such events are made more sustainable every year, our team wants to make these events not only based on revenue, but also sustainability of the city held as one of the core factors to consider in planning. With this thought, we're hired by **Consulting Organization Making Athletics Planet positive (COMAP)**, we created both the **City Impact Evaluation Model (CIEM)** and **National Impact Evaluation model (NIEM)** that assess the environmental footprint of the host cities and countries for the Super Bowl and the World Cup, respectively, striving for greener, more responsible sporting events in the years to come.

1.2 Problem Restatement

In this study, we aim to complete the following tasks.

- **Task 1:** To analyze the environmental footprints of a specific event (**Super Bowl LIX**) and to identify factors that impact the environmental footprint of this mega sporting event by referring to **existing metrics**.
- **Task 2:** To create a **qualitative or quantitative model** to determine where to hold the next Super Bowl based **solely** on environmental concerns.
- **Task 3:** To **apply** our model in **Task 2** to determine (1) which of the previously hosting cities—Inglewood (CA), Glendale (AZ), Las Vegas (NV), and New Orleans (LA)—should host the Super Bowl again, and (2) which of the three selected cities that have never hosted before represents the most sustainable and environmentally friendly choice.
- **Task 4:** To **adapt** our model to assess the environmental footprint of a single-sport championship tournament (**the World Cup**) and to determine the best host location while accounting for the event's unique environmental challenges; to develop **stratagems** that minimize environmental impact during the event (short term) and are economically and logistically practical (long term); to compare the **key environmental factors** between the Super Bowl and the single-sport championship tournament (the World Cup) and to provide explanations.
- **Task 5** To write a one-page persuasive **letter** to the NFL committee recommending our chosen city for Super Bowl LXIII in 2029, emphasizing the environmental advantages of the city and its alignment with the NFL's sustainability goals.

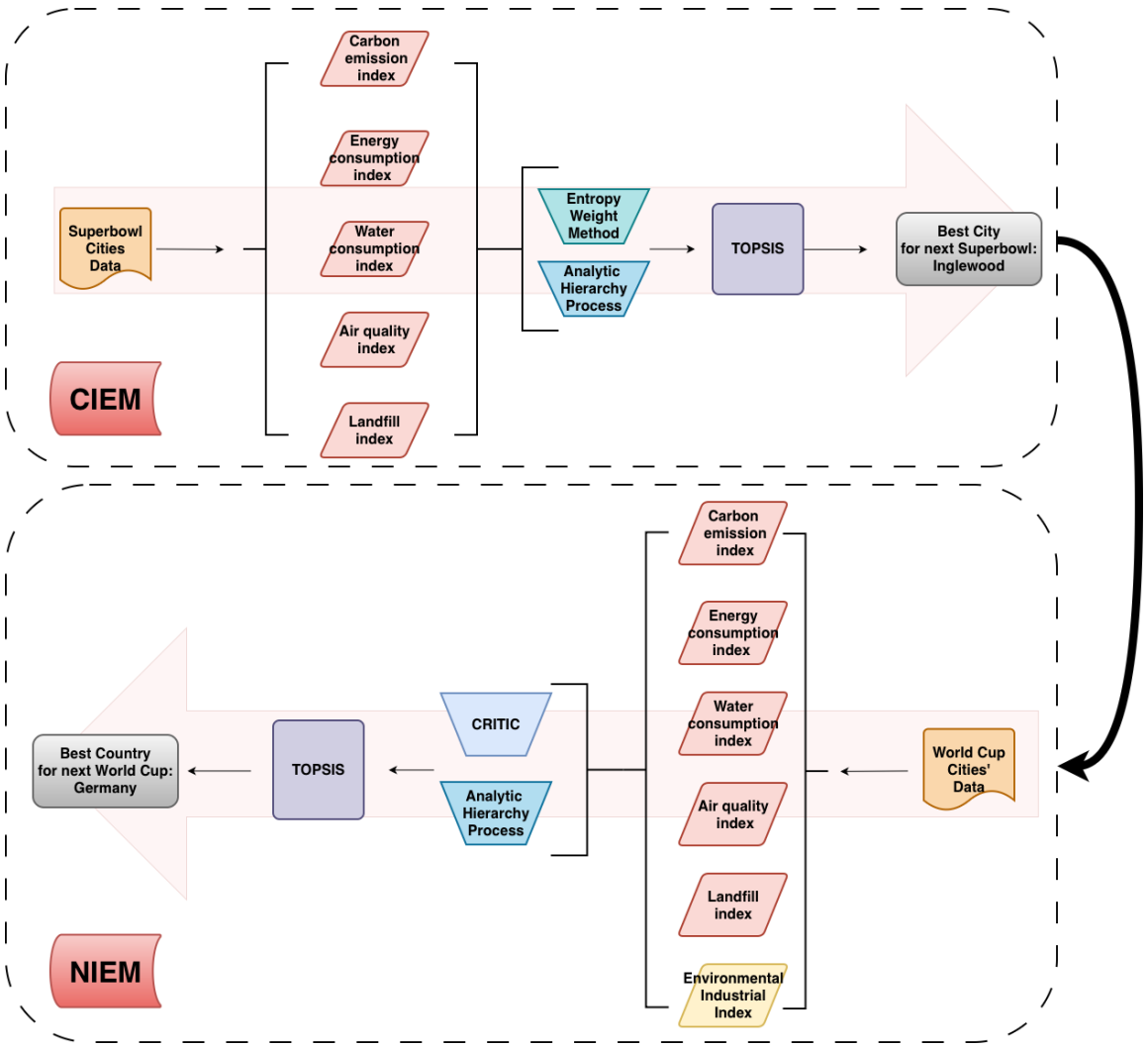


Figure 1.1: Flowchart of our study

2 Assumptions with Justification

Assumption 1: The scale and duration of the Super Bowl are constant in all cities and all times. We assume that there are **80000 participants** (including athletes and audiences) and a period of 3 days for simplicity[1].
Justification: This makes sure that our comparison between cities is fair and consistent and that the differences in the model's output are due to characteristics of the host city rather than variations in the event's scale or duration. To clarify, the number of audiences for one Super Bowl event is about 70000 people, according to our research. Since every Super Bowl event has about the same number of audience, we simply set the maximum people in a Super Bowl event as 80000 people due to other faculty members and participants.

Assumption 2: The behavior of Super Bowl participants is consistent in all cities and events.
Justification: This makes sure the effect of the city's infrastructures and the calculations are not affected by multifarious personal situations.

Assumption 3: State-level data serves as proxies for city-level environmental characteristics.
Justification: While some variation exists within states, state-level data is considered to be comprehensive and standardized, which can reflect the overall environmental characteristics of the host city.

Assumption 4: Environmental policies remain stable from 2020 to 2029, and all urban infrastructure systems function normally without disruptions during the event.
Justification: This ensures that the model assesses cities based on their existing environmental governance and typical infrastructures, rather than being influenced by unpredictable policy changes or infrastructure failures.

Assumption 5: The number of audiences for a World Cup event is about 3.5×10^6[5]. The number of audiences per venue is about 6×10^4. We assume that this is the case for all World Cup events.
Justification: These estimations are based on historical attendance data and therefore reflect a typical number of participants during a World Cup event. Moreover, while the actual number may deviate slightly from the assumed value, these estimations allows us to yield practical evaluation for different host countries.

Assumption 6: Super Bowl LIX held in New Orleans generated a considerably low amount of landfill waste, making it suitable to be an ideal reference when calculating the Landfill Index (LFI).
Justification: Super Bowl LIX took several measures to minimize landfill build-up. Indeed, it aims to divert 'at least 90% of waste from landfills'[6].

3 Environmental Evaluation of Super Bowl LIX

3.1 Establishment of an Environmental Footprint Framework

Through thorough research, we have dissected the environmental impact of Super Bowl LIX into five core components: **Carbon emissions**, **energy consumption**, **water impact**, **air quality**, and **landfill**. Notably, these components are each based on existing metrics or evaluations.

3.2 Quantitative Impact Analysis by Category

- **Carbon Emissions:** Carbon emissions are arguably one of the major concerns when it comes to environmental impact assessment. BioMed Central (BMC), a global authority in scientific publishing, defines the metric carbon footprint reduction index (CaFRI) as "a comprehensive greenness assessment tool prioritizing carbon footprint as the primary environmental impact"[9], highlighting the significance of measuring carbon emission. Throughout our research, we roughly estimated the carbon emissions in New Orleans by looking at carbon emissions per capita in Louisiana[4]. Since there are 65,719 offline audiences, the carbon dioxide emission of Super Bowl LIX was **about 21,660 tons**[18].

- **Energy Consumption:** Energy consumption, especially electricity consumption, "is at the heart of modern economies" [8], meaning that energy consumption is also a key factor that influences the environmental impact of Super Bowl LIX. Looking at energy consumption per capita in Louisiana[3], we roughly estimated the total energy consumption to be **491,176,000,000 British thermal units (Btu)**

- **Water Footprint:** Water consumption also plays a crucial role when it comes to assessing the environmental impact of mega sporting events. "Outdoor sports facilities use millions of gallons of water, putting pressure on local water supplies, as well as using hazardous pesticides can pollute soil and contaminate run-off into the water, threatening employees, players, and animals"[11]. Due to the scarcity of online data resources, we first used AI to conduct the estimation (we will utilize a different way to estimate when constructing our model). Considering drinking and restroom usage of the participants during Super Bowl LIX, we roughly estimated the water consumption to be about **4 million litres**.

- **Air Quality:** Air quality and pollution are considerably influential when assessing mega sporting events. According to a study on Super Bowl, "significantly higher levels of pollution are observed on Super Bowl Sunday"[12][17] "millions dying each year from exposure to and the side effects of air pollutants". Based on a specific study towards the AQI caused by NFL stadiums. It points out that ozone increases significantly with more audiences. According to a study, the maximum AQI of an NFL event reaches **187**, which is considerably severe for the local environment[17].

- **Landfill:** The immense waste generated by mega-sporting events like the Super Bowl has a large impact on the environment. Landfills not only cause havoc on natural ecosystems, leading to water contamination, soil pollution, and air pollution, it also leads to the emission of methane, a greenhouse gas. By research, around **2,000 tons** of food waste is generated at Super Bowl parties. Even though the proportion of landfill for Super Bowl LIX is just below 10%, this number is still considerably large for the amount of total trash produced for the state[20].

4 City Impact Evaluation Model (CIEM): Indices

Symbol	Description	Unit
A_i	State area	km^2
AQI_i	Air quality index in each state	N/A
c_i	State carbon emission per capita (per day)	tCO_2e
e_i	Average residential energy consumption per capita (per day)	Btu
L	Total landfill generated by a single Super Bowl event	kg
p_i	State population	N/A
p_t	Total US population	N/A
r_i	Proportion of renewable energy (vs non-renewable)	%
t_i	Temperature of the host state/city in February	$^\circ F$
T_i^c	Carbon emission by transportation per capita (per day)	tCO_2e
T_i^e	Average energy consumption by transportation per capita (per day)	Btu
W_{ci}	Water consumption per capita in the host city	t
W_s	Total US water supply	t

Table 4.1: Symbols, Descriptions, and Units

To perform a comprehensive evaluation for each potential host city, we established 5 key indices that measure 5 core aspects of environmental impact: **Carbon Emission Index (CEI), Energy Consumption Index (ECI), Water Consumption Index (WCI), Air Quality Index (AQI), and Landfill Index (LFI)**. Notably, all 5 indices are **cost-type** (i.e., smaller values are more preferable).

4.1 Carbon Emission Index (CEI)

Carbon Emission Index (CEI) indicates the impact of carbon emission to the state. According to research, transportation accounts for 15% of anthropogenic greenhouse gas emissions[7]. Therefore, other than **carbon emission per capita**, we decided to add the effect of **transportation** to the calculation of CEI as well. Hence, CEI is calculated by the following formula:

$$I_{Ci} = \frac{(\frac{|t_i - t_{comfort}|}{t_{max} - t_{min}} + 1) \times 80000 \times c_i + \epsilon \times 80000 \times T_i^c}{(p_i + 80000)/A_i} \tag{4.1}$$

Where I_C is the **CEI**.

Notably, t_i is the temperature of the state in February; $t_{comfort}$ is a constant that represents the most comfortable temperature for human body and has the value of 77 degrees Fahrenheit[10]; t_{max} and t_{min} refer to the maximum and minimum temperature in February, respectively. Since greater deviation from the comfort temperature of 77 degrees F leads to **the use of air conditioning or heating systems that release carbon dioxide either directly or indirectly**, temperature also influences CEI of a city (e.g., greater difference between actual temperature t_i and the most comfortable temperature $t_{comfort}$ results in greater carbon emission). The addition of 1 prevents I_C from becoming 0 because of the city having the most comfortable temperature.

Additionally, p_i refers to the population of the state, and c_i represents carbon emission by residents per day. The product of the two quantities aforesaid indicates, thus, the **total emission of carbon by residents**.

Beyond that, ϵ is the quotient of the number of transportation divided by the number of audience, which is essentially a **per-capita transport intensity** (i.e., the number of transport units required per audience member). Thus, the product of ϵ, number of participants (80000), and average carbon emission by transportation per capita (T_i^c) gives us the overall transport-related carbon footprint of all audiences.

To obtain the coefficient of ϵ, we took the reciprocal of the average vehicle occupancy rate, which is about 3.0. Consequently:

$$\epsilon \approx 0.33$$

Together, the numerator of the fraction on the right hand side represents the total carbon emission by the event. Dividing the numerator by population density (**total number of people within the state**, which is the sum of the state population, p_i, and the number of participants, 80000, over the **area of the state**, the fraction would be **spatial carbon intensity per person** (i.e., how much carbon emission is associated with each unit area per person).

To **normalize** the **CEI** and obtain S_1, we subtract the sample minimum from I_C and divide the difference aforesaid by the interquartile range, expressed as follows:

$$S_1 = \frac{I_{Ci} - I_{Cmin}}{IQR_C} \tag{4.2}$$

4.2 Energy Consumption Index (ECI)

In order to measure the impact of Super Bowl on local areas, we looked at two types of energy consumption: annual energy consumption per capita and average energy consumption by transportation per capita. Additionally, we looked at the use of renewable energy in each state. The equation that calculates the **Energy Consumption Index (ECI)**, which means the impact of energy consumption, is as follows:

$$I_{Ei} = \frac{80000 \times e_i \times \beta + \epsilon \times 80000 \times T_i^e}{r_i} \tag{4.3}$$

Here, e_i is the annual residential energy consumption per capita, and β is a coefficient that measures how much more energy each person consumes during the Super Bowl session. Estimating based on data from previous major sporting events:

$$\beta \approx 0.46$$

Therefore, the product of the number of participants (80,000), e_i, and β gives us **the total residential energy consumption** by the participants.

Similar to T_i^c in the carbon emission index, T_i^e refers to the average energy consumption by transportation per capita. ϵ, as mentioned above, is the per-capita transport intensity. Thus, the product of ϵ, the number of participants (80,000), and T_i^c gives **the total energy consumption from transportation** by the participants.

r_i represents the proportion of renewable energy within the state relative to non-renewable energy. By making r_i the denominator, we aim to convey the idea that it is a benefit-type indicator. The greater the value of r_i, the smaller the value of I_E, indicating a more sustainable use of energy.

Again, normalizing I_E, we obtained S_2:

$$S_2 = \frac{I_{Ei} - I_{Emin}}{IQR_E} \tag{4.4}$$

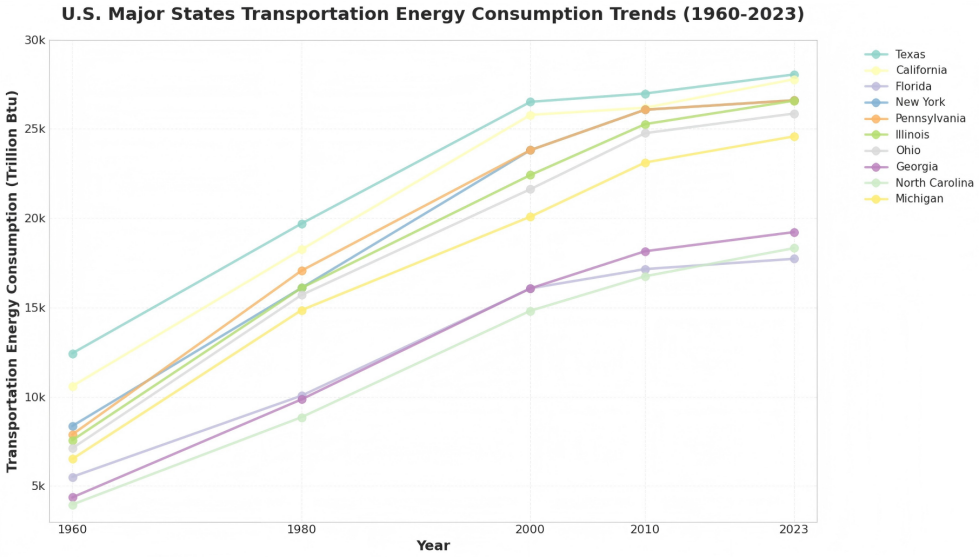


Figure 4.1: Energy cost due to transportation in the US

The trend of transportation consumption in each state indicates that the increment of energy consumption due to transportation is decreasing with time. Therefore, we take **the logarithm of the value** for transportation of energy consumption to perform the linear regression.

4.3 Water Consumption Index (WCI)

Formulating I_W, the **Water Consumption Index (WCI)**, we incorporated three core factors into our equation:

$$I_{Wi} = \frac{(80000 \times \gamma + p_i)W_{ci}}{logA_i} \times \frac{1}{W_s \times \frac{p_i}{p_t}} \times \frac{1}{p_i \times W_{ci}} \tag{4.5}$$

In a nutshell, we considered three core factors that influence the **WCI**: the total water consumption **during a Super Bowl event** per unit area within the state (the first fraction), the total water supply by the state (the second fraction), and the **usual** water consumption within the state (the third fraction).

First, humans, irrespective of whether they are athletes or audiences, will **consume more water** during a Super Bowl event. To elaborate, athletes consume more water because of intense physical activities, while audiences consume slightly more water because of emotional arousal (such as excitement, tension, or cheering). Interestingly, according to our research, athletes will consume **about 2 to 3 times more** water than usual during exercise, while audiences will consume **about 1.33 times more** water than usual when watching a sport[2][13]. Therefore, we established the constant γ, representing **a water-consumption multiplier** that shows how many times more water, on average, a participant uses during the event compared to their normal usage. Based on the information aforesaid about additional water consumption during a sporting event, we calculated gamma to be:

$$\gamma \approx 1.35$$

The product of the number of participants (80000) and γ is thus the **total water need by the sporting event**. This sum of this value and the state population (p_i) gives the total water need within the state. Multiplying the total water need by state water consumption per capita (W_{ci}), subsequently, gives the total water consumption within the state.

As the area of each state is too vast and therefore is **inconvenient for calculation**, we take the logarithm of the state area ($logA_i$). Together, total water consumption within the state over the area gives the **total water consumption during a Super Bowl event per unit area within the state**.

Second, we were unable to procure any data about the water supply within each state. Solving this issue, we used a fraction of the state population over the US population ($\frac{p_i}{p_t}$) to **estimate** the water supply needed by each

state. Multiplying the total water supply within the US (W_s) by the fraction, consequently estimates the **water supply within each state**. This is the denominator, which shows that $W_s \times \frac{p_i}{p_t}$ is a **cost-type** variable (i.e., the greater the water supply within the state, the smaller the WCI, which is a cost-type index as mentioned above)

Third, the product of the state population (p_i) and the state water consumption per capita (W_{ci}) gives the **total water consumption within the state**. $p_i \times W_{ci}$ is a **benefit-type variable**, as states with higher baseline water consumption would be less impacted by additional water use during a Super Bowl event. Therefore, $p_i \times W_{ci}$ is set to be the denominator (i.e., the greater the total water consumption within the state, the smaller the WCI).

Interestingly, the W_{ci} **can be directly canceled out** from the equation, as state water consumption per capita data is not available online.

Similarly, we obtained our third index (S_3) by normalizing **WCI** with the same method:

$$S_3 = \frac{I_{Wi} - I_{Wmin}}{IQR_W} \tag{4.6}$$

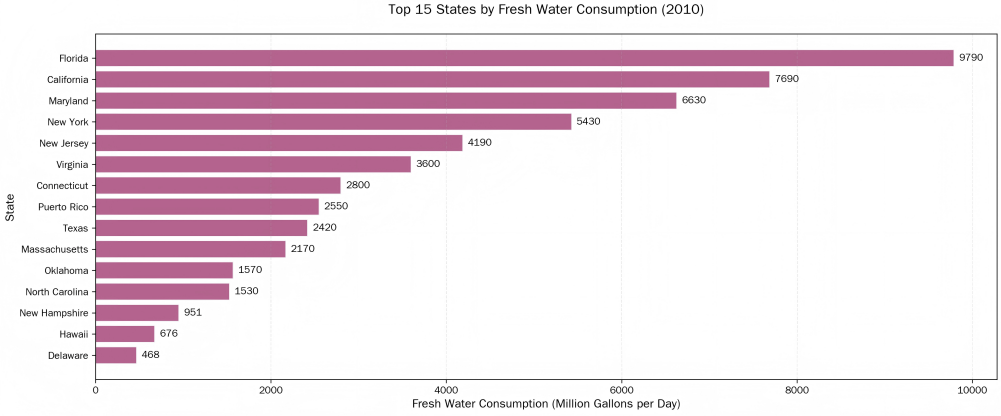


Figure 4.2: Water consumption in the US, 2010[15]

4.4 Air Quality Index (AQI)

AQI is an index that picks the **pollutant with largest impact** to the air quality. The formula for calculating AQI is:

$$AQI = max(IAQI_1,\ IAQI_2,\ IAQI_3,\ IAQI_4,\ ...) \tag{4.7}$$

Where $IAQI_i$ stands for the air quality index of individual pollutants; the formula for $IAQI_i$ is:

$$IAQI_i = \frac{IAQI_{Hi} - IAQI_{Lo}}{BP_{Hi} - BP_{Lo}} \times (C_i - BP_{Lo}) + IAQI_{Lo} \tag{4.8}$$

Here, C_i is the concentration measurement value of pollutant i; BP_{Hi} is the concentration limit higher than C_i; BP_{Lo} is the concentration limit lower than C_i in the chart; $IAQI_{Hi}$ is the $IAQI$ value corresponding to BP_{Hi};$IAQI_{Lo}$ is the $IAQI$ value corresponding to BP_{Lo}.

Here, we picked two substances with the most impact on AQI: **ozone and PM2.5**

Notably, the overall trend of **PM2.5 is decreasing**, and the trend of **ozone level remains constant**. Therefore, the ozone level is **presumably the highest** compared to other air pollutant levels, including the level of PM2.5, in 2029, and therefore we will use the predicted ozone level in 2029 to calculate the AQI for each potential host city[17].

Finally, we used the same method to normalize the AQI:

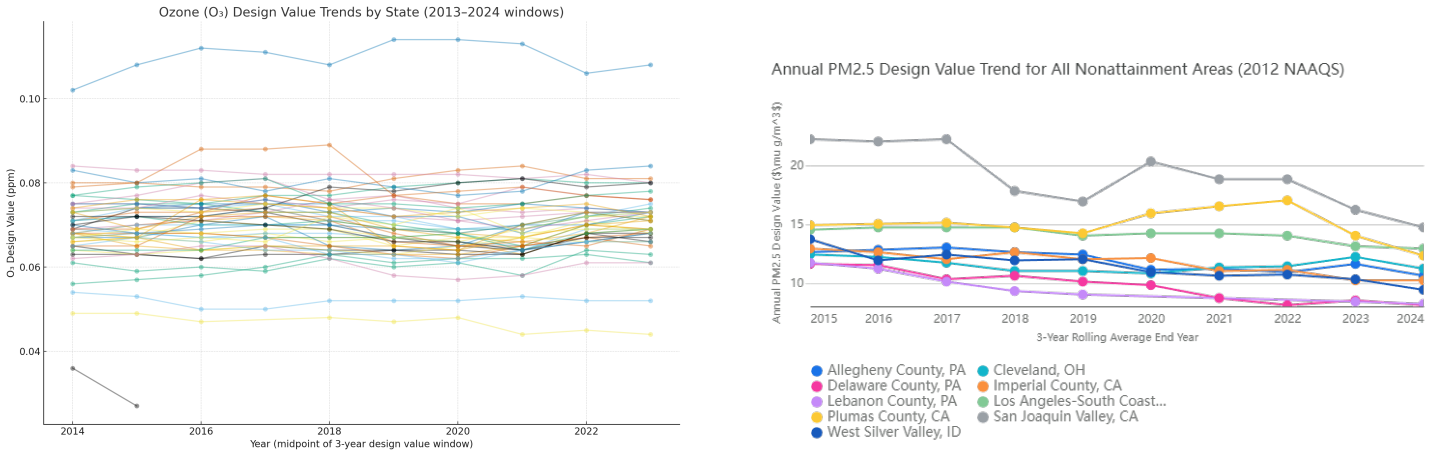


Figure 4.3: Ozone and PM2.5 Design value trends

$$S_4 = \frac{AQIi - AQI_{min}}{IQR_{AQI}} \tag{4.9}$$

4.5 Landfill Index (LFI)

To assess the landfill or waste generated by a Super Bowl event in a potential host city, we first looked at **the amount of waste** generated by a Super Bowl event and the state area. Additionally, we took Super Bowl LIX in New Orleans, which has a relatively low landfill, as a reference.

$$I_{Ri} = \frac{L}{logA_i} \times (e_i^{th} - {}^{N}e^{th})^2 \tag{4.10}$$

Here, L refers to the total landfill generated by a Super Bowl event, A the state area. Similar to WCI, we took the logarithm of the state area (A_i), as the area of each state is too vast and therefore inconvenient for calculation. Together, the first fraction represents the **landfill or waste per unit area**. Again, this is a cost-type variable (i.e., the greater the landfill per unit area, the greater the LFI, which is a cost-type index).

Moreover, e^{th} represents the **environmental quality index**, with a low e^{th} representing a higher environmental quality, ranging from 0 to 1. By measuring the **distance** of the **environmental quality index** of other potential host cities from **that of Super Bowl LIX in New Orleans** (${}^{N}e^{th}$), we obtained a relative assessment of them.

Again, we employed the same method for normalization:

$$S_5 = \frac{I_{Ri} - I_{Rmin}}{IQR_R} \tag{4.11}$$

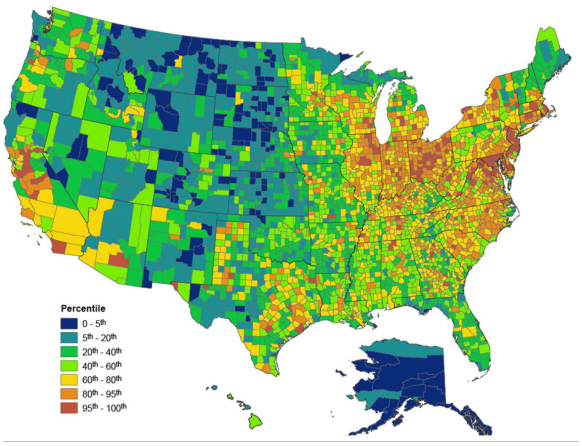


Figure 4.4: Overall Environmental Quality Index by County 2006-2010[14]

On the graph, it is shown clearly how the environment varies across the US, demonstrating that e^{th} is a significant component of LFI.

4.6 Computed City Indices Values

Linear Regression Fitting Process

n data points are given as (x_i, y_i), where $i = 1, 2, \ldots, n$.

1. Compute Means

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \quad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i \tag{4.12}$$

2. Compute Sums of Squares

$$S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2, \quad S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) \tag{4.13}$$

3. Calculate Regression Coefficients

$$b = \frac{S_{xy}}{S_{xx}}, \quad a = \bar{y} - b\bar{x} \tag{4.14}$$

4. Obtain Fitted Linear Equation

$$y = a + bx \tag{4.15}$$

5. Prediction for Future Value For a future x_{future}:

$$y_{\text{predicted}} = a + b \cdot x_{\text{future}} \tag{4.16}$$

Values Exhibition

After predicting the required data for 2029, we obtained the five indices (S_1 to S_5) for each of the seven cities. For each index, the values **across the seven cities were normalized within that index**. To make all 35 values **directly comparable**, nonetheless, we need to perform a global normalization across all indices and cities. The globally normalized data are as follows:

Table 4.2: Normalized Values for Selected Cities

	Inglewood	Glendale	Las Vegas	New Orleans	Nashville	Charlotte	Chicago
S_1	0	0.036	0.178	0.416	0.128	0.059	0.453
S_2	0	0.295	0.098	0.498	0.345	0.178	0.095
S_3	0	0.395	0.792	0.975	1	0.692	0.555
S_4	0.702	0.372	0.132	0.122	0.145	0	0.278
S_5	0.228	0	0.236	0	0.258	0.884	0.25

5 City Impact Evaluation Model (CIEM): Weighting and Evaluation

Four cities that have hosted, which include **Inglewood (CA), Glendale (AZ), Las Vegas (NV), and New Orleans (LA)**, and three NFL cities that have never hosted, which include **Nashville (TN), Charlotte (NC), and Chicago (IL)**, were chosen to be assessed.

After establishing and calculating the 5 indices (the 5 variables) of the 7 cities, we determined the weights of each by a strategic combination of Analytic Hierarchy Process (AHP) and Entropy Weight Method (EWM).

5.1 Subjective Weighting: Analytic Hierarchy Process (AHP)

The AHP decomposes the decision problem into a hierarchy and evaluates subjectively through **pairwise comparisons**, thereby calculating the weights of each indicator.

Judgment Matrix Construction

Based on our subjective evaluations, we constructed the following judgment matrix for the five environmental indicators using the standard 1-9 scale:

$$A = \begin{bmatrix} 1 & 3 & 5 & 7 & 9 \\ 1/3 & 1 & 3 & 5 & 7 \\ 1/5 & 1/3 & 1 & 3 & 5 \\ 1/7 & 1/5 & 1/3 & 1 & 3 \\ 1/9 & 1/7 & 1/5 & 1/3 & 1 \end{bmatrix} \tag{5.1}$$

Indicator order: S1, S2, S3, S4, S5

Rationale of the judgment matrix:

Carbon emissions are weighted highest due to their **global climate impact and long-term irreversible effects**. Energy consumption follows as it directly drives carbon emissions. Water resources rank third because it is more regional, rather than global. While air quality and landfill management address more localized environmental concerns, so they are assigned with least importance.

Weight Calculation

The normalized weights are calculated using the geometric mean method:

$$w_i = \frac{M_i}{\sum_i M_i}, \quad \text{where } M_i = \left(\prod_{j=1}^{5} a_{ij} \right)^{1/5} \tag{5.2}$$

The resulting set of weights is as follows:

$$W_s = [0.509, 0.264, 0.130, 0.064, 0.033]^T \tag{5.3}$$

Verification: $0.509 + 0.264 + 0.130 + 0.064 + 0.033 = 1.000$

Consistency Check

Maximum eigenvalue: $\lambda_{\text{max}} = 5.141$

Consistency Index: $CI = 0.036$

Consistency Ratio: $CR = \frac{CI}{RI} = 0.031$ (where RI = 1.12 for n=5)

Since $CR = 0.031 < 0.10$, the judgment matrix passes the consistency check.

5.2 Objective Weighting: Entropy Weight Method (EWM)

The EWM determines weights based on the information content of the data, which is objective. An indicator with greater data variability is assigned a higher weight.

Data Preparation and Normalization

Using the normalized scores S_1 to S_5 and applying cost-type indicator normalization:

$$r_{ij} = \frac{\max(x_j) - x_{ij}}{\max(x_j) - \min(x_j)} \tag{5.4}$$

Entropy Weight Calculation

The normalized **benefit-type matrix** R is first converted to a proportion matrix P, where each element p_{ij} represents the proportion of the j-th indicator for the i-th city:

$$p_{ij} = \frac{r_{ij}}{\sum_{i=1}^{7} r_{ij}} \tag{5.5}$$

The entropy weights are then calculated as follows:

$$w_j = \frac{1 - e_j}{\sum_{j=1}^{5}(1 - e_j)}, \quad \text{where } e_j = -k \sum_{i=1}^{7} p_{ij} \ln(p_{ij}), \quad k = 1/\ln(7) \tag{5.6}$$

Final EWM weights: $W_o = [0.224, 0.159, 0.215, 0.148, 0.253]^T$

Verification: $0.224 + 0.159 + 0.215 + 0.148 + 0.253 = 0.999 \approx 1.000$

EWM Calculation Results

Table 5.1: Entropy Weights Calculation Results

Indicator	**Entropy Value** (e_j)	**Difference** ($1 - e_j$)	**Entropy Weight** (w_j)	**Rank**
Carbon Emission Index (S_1)	0.782	0.218	0.224	2
Energy Consumption Index (S_2)	0.845	0.155	0.159	4
Water Consumption Index (S_3)	0.791	0.209	0.215	3
AQI Index (S_4)	0.856	0.144	0.148	5
Landfill Index (S_5)	0.754	0.246	0.253	1
Total	4.028	0.972	**0.999**	

Strategic AHP-EWM Combination

We combine the **subjective AHP weights** (W_s) and **objective EWM weights** (W_o) in a linear fashion:

$$W_c = \alpha W_s + (1 - \alpha) W_o, \quad \alpha = 0.6 \tag{5.7}$$

We set the value of α to be 0.6 (60%), meaning that W_s, the subjective AHP, is slightly more important than W_o, the objective EWM, because we deemed our research- and experience-based insights to have greater practical value.

Table 5.2: Final Combined Weights Calculation

Indicator	**AHP Weight** (W_s)	**EWM Weight** (W_o)	**Combined Weight** (W_c)	**Rank**
CEI (S_1)	0.509	0.224	0.390	1
ECI (S_2)	0.264	0.159	0.222	2
WCI (S_3	0.130	0.215	0.164	3
AQI (S_4))	0.064	0.148	0.098	5
LFI (S_5)	0.033	0.253	0.126	4
Total	**1.000**	**0.999**	**1.000**	

Verification of Combined Weights: $0.390 + 0.222 + 0.164 + 0.098 + 0.126 = 1.000$

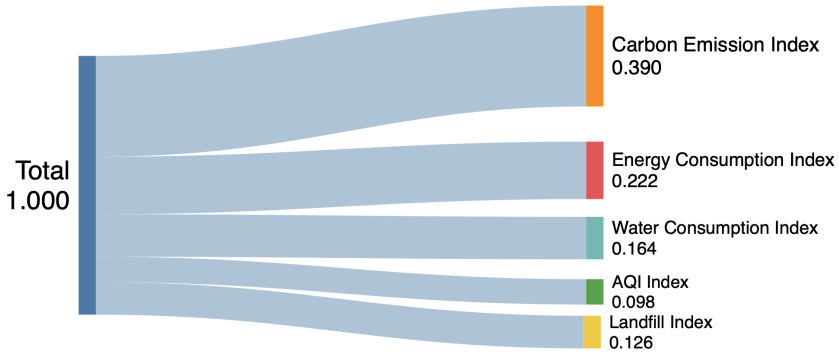


Figure 5.1: Sankey Diagram of Weight Distribution across Methods

The final weights provide a balanced assessment that incorporates both subjective judgment from AHP and data-driven calculation from EWM, with CEI (0.390) and ECI (0.222) being the most important factors for environmental sustainability evaluation.

5.3 Comprehensive Evaluation: Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)

The Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) was employed to holistically assess the potential host cities based on the five key indices.

First, we performed **vector normalization** for the 35 values (7 cities, 5 indices for each), with the 35 new values calculated by the following equation:

$$x_{ij}^* = \frac{x_{ij}}{\sqrt{\sum_i x_{ij}^2}} \tag{5.8}$$

Then, in addition to our established weights, we established our **Positive Ideal Solution (PIS)** and **Negative Ideal Solution (NIS)**:

$$w_j = (w_1, w_2, w_3, w_4, w_5) = (0.390, 0.222, 0.164, 0.098, 0.126)$$
$$PIS = (a_1, a_2, a_3, a_4, a_5) = (0, 0, 0, 0, 0)$$
$$NIS = (b_1, b_2, b_3, b_4, b_5) = (0.689539, 0.701056, 0.532829, 0.804313, 0.876241)$$

The score of each city, subsequently, was calculated by:

$$s_i = \frac{\sqrt{\sum_j (x_{ij}^* w_j - b_j)^2}}{\sqrt{\sum_j (x_{ij}^* w_j - b_j)^2} + \sqrt{\sum_j (x_{ij}^* w_j - a_j)^2}} \tag{5.9}$$

5.4 Results

The results are shown in **Table 5.3**: This shows that Inglewood, California, is the most suitable location to host Super Bowl LXIII due to its leading TOPSIS score of 0.7038 out of all seven cities and its exceptional performance in **carbon emission, energy use, and water consumption**.

6 Sensitivity Analysis

6.1 Weight parameter α sensitivity analysis

The weight parameter α balances the subjective AHP weights and objective EWM weights in our weight combination process. The combined weights are given by:

Table 5.3: Environmental Impact Scores of Selected Cities

City	Score
Inglewood (CA)	0.7038
Glendale (AZ)	0.6842
Las Vegas (NV)	0.6446
Charlotte (NC)	0.5770
Nashville (TN)	0.5551
Chicago (IL)	0.4545
New Orleans (LA)	0.4021

$$W_c = \alpha W_s + (1 - \alpha) W_o \tag{6.1}$$

where $\alpha \in [0, 1]$. A higher α emphasizes AHP, while a lower α emphasizes EWM.

Variation of α

We calculate the combined weights for different values of α: 0.5, 0.6, 0.7, and 0.8. Using the previously calculated W_s and W_o, the combined weights W_c are as follows:

Table 6.1: Combined Weights for Different α Values

Indicator	$\alpha = 0.5$	$\alpha = 0.6$	$\alpha = 0.7$	$\alpha = 0.8$
Carbon Emission Index	0.367	0.390	0.413	0.436
Energy Consumption Index	0.212	0.222	0.232	0.242
Water Consumption Index	0.173	0.164	0.156	0.148
AQI Index	0.106	0.098	0.090	0.082
Landfill Index	0.143	0.126	0.109	0.092
Total	1.001	1.000	1.000	1.000

Impact on City Scores and Rankings

We recomputed the TOPSIS scores for each α value using the corresponding combined weights:

Table 6.2: TOPSIS Scores and Rankings for Different α Values

City	$\alpha = 0.5$		$\alpha = 0.6$		$\alpha = 0.7$		$\alpha = 0.8$	
	Score	**Rank**	**Score**	**Rank**	**Score**	**Rank**	**Score**	**Rank**
Inglewood (CA)	0.695	1	0.704	1	0.712	1	0.720	1
Glendale (AZ)	0.677	2	0.684	2	0.691	2	0.698	2
Las Vegas (NV)	0.649	3	0.645	3	0.640	3	0.635	3
Charlotte (NC)	0.580	4	0.577	4	0.574	4	0.571	4
Nashville (TN)	0.558	5	0.555	5	0.552	5	0.549	5
Chicago (IL)	0.458	6	0.455	6	0.451	6	0.447	6
New Orleans (LA)	0.405	7	0.402	7	0.398	7	0.394	7

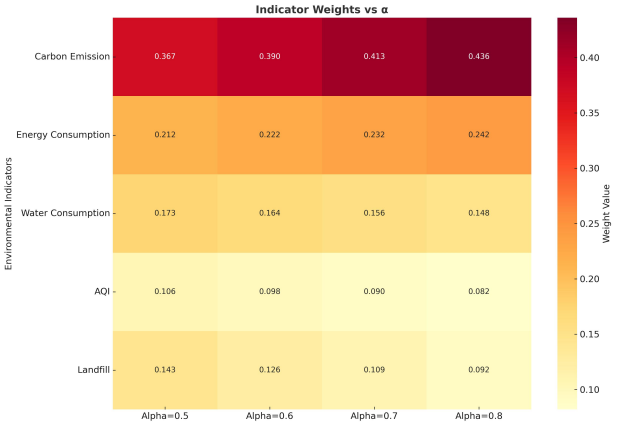


Figure 6.1: Weights heatmap

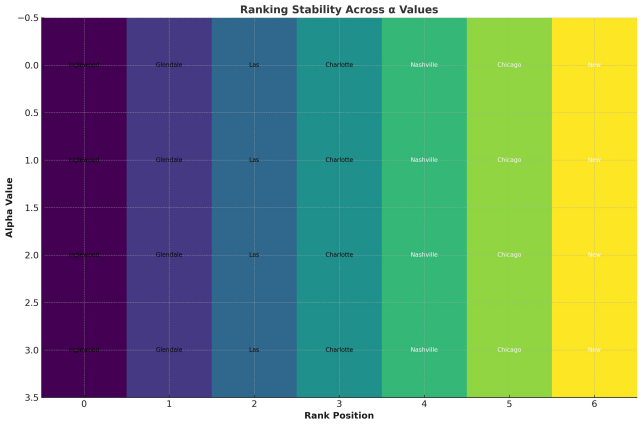


Figure 6.2: Ranking stability

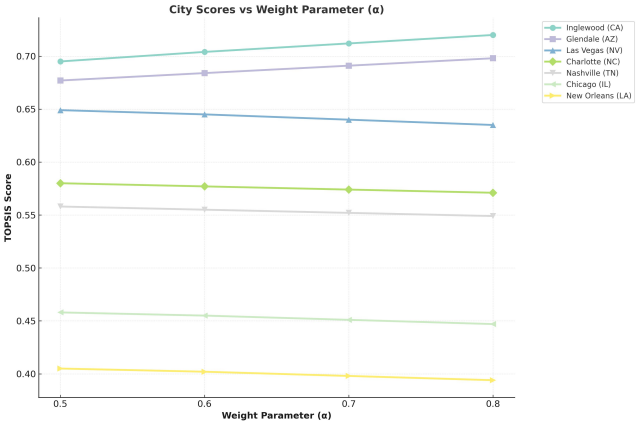


Figure 6.3: City Score Trends Across Different Alpha Values

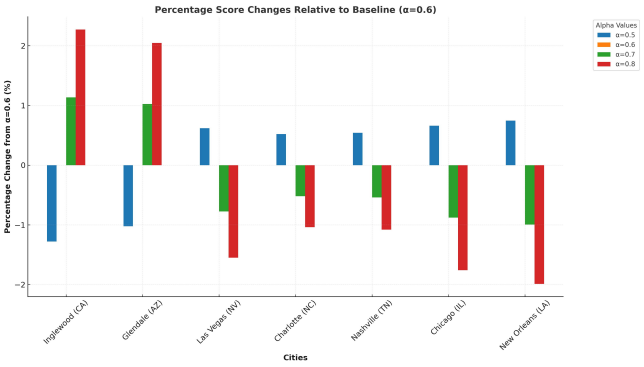


Figure 6.4: Percentage Score Changes Relative to Baseline

Analysis of Sensitivity Results

Figure 6.1 and in Figure 6.3 indicates and illustrates that as α increases, AHP weights more in the combined weight, which cities with better performance on the high-priority AHP indicators (Inglewood and Glandale) show slight score improvements, while others show a slight decrease in score. As shown in Figure 6.4, the maximum score variation between different values α is 0.016 (Inglewood), representing only 3.6% relative change from baseline. The relative ranking of all seven cities remains completely unchanged across all α values, as shown in Figure 6.2.

6.2 Number of Participants Sensitivity Analysis

To further validate the robustness of our model, we conducted sensitivity analysis on the "Participant Number" parameter, which was initially set at 80,000. We tested variations of ±10% to assess how changes in attendance affect the environmental impact scores and city rankings.

Methodology

We modified the audience parameter N in our environmental impact formulas and the key formulas affected by this parameter are:

- Carbon Emission Index: $I_C \propto N \times c_i + \epsilon \times N \times T_i^c$
- Energy Consumption Index: $I_{Ei} \propto N \times e_i \times \beta + \epsilon \times N \times T_i^e$
- Water Consumption Index: $I_W \propto N \times \gamma + p_i$

Results

Table 6.3: City Scores under Different Audience Scenarios

City	-10% (72,000)	Base (80,000)	+10% (88,000)	Rank (-10%)	Rank (Base)	Rank (+10%)
Inglewood (CA)	0.7012	0.7038	0.7063	1	1	1
Glendale (AZ)	0.6815	0.6842	0.6868	2	2	2
Las Vegas (NV)	0.6421	0.6446	0.6470	3	3	3
Charlotte (NC)	0.5753	0.5770	0.5786	4	4	4
Nashville (TN)	0.5532	0.5551	0.5569	5	5	5
Chicago (IL)	0.4528	0.4545	0.4561	6	6	6
New Orleans (LA)	0.4004	0.4021	0.4037	7	7	7

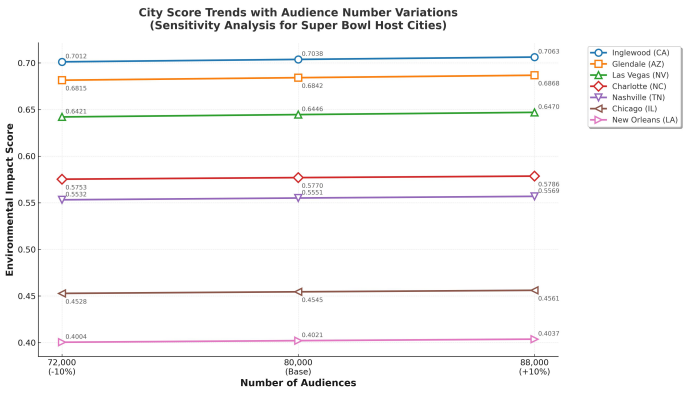


Figure 6.5: City Score Trends

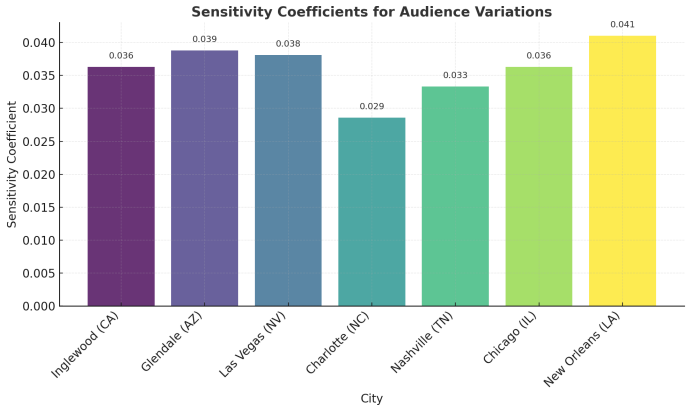


Figure 6.6: Sensitivity Coefficients

Analysis of Sensitivity Results

The relative ranking remains completely unchanged, demonstrating high robustness. The maximum absolute variation is 0.0051 (Inglewood), representing only 0.72% relative change from baseline. All sensitivity coefficients are below 4.1% (average 3.6%), confirms model stability.

Sensitivity Analysis Conclusion

This shows the stability of our results under weight variations and participant number variation, which provides strong evidence for the reliability of our host city recommendations. Table 6.2 and Table 6.3 clearly shows that Inglewood maintains its top position across all weighting scenarios, reinforcing our confidence in selecting it as the most environmentally sustainable choice for Super Bowl 2029.

7 National Impact Evaluation Model (NIEM)

Overview

The **National Impact Evaluation Model (NIEM)** evaluates the environmental impact of potential host countries for the **World Cup**, a single-sport championship tournament.

The World Cup is usually hosted by a single country, which is arguably similar to the Super Bowl. Unlike the Super Bowl, which is a single-day event in a specific stadium, nevertheless, the World Cup is a **month-long** tournament held in several **different places within the country**.

Therefore, to assess the environmental sustainability of potential host cities for the World Cup, we expanded the City Impact Evaluation Model (CIEM) and developed National Impact Evaluation Model (NIEM), a specialized evaluation model for the **World Cup** host countries.

When applying our model to potential host countries, we decided to pick 7 countries that have hosted the World Cup before: **Qatar, Russia, Brazil, The Republic of South Africa, Germany, South Korea, and Japan**.

Notably, NIEM is essentially an extended CIEM. Specifically, we adjusted the 5 indices from CIEM and added a new index. Afterwards, instead of utilizing the combination of Analytical Hierarchy Process (AHP) and Entropy Weight Method (EWM), we melded AHP with **CRiteria Importance Through Intercriteria Correlation (CRITIC)** to better handle the textbfinterdependence between the indices while still ensuring objectivity within subjectivity. Subsequently, we employed **TOPSIS** again to yield a comprehensive rating for each of the potential host countries.

7.1 Addition and Adjustments of Indices

Added: Environmental-Industrial Index (EII)

Different countries have varying degrees of industrialization, meaning they have different potentials in handling environmental issues. In CIEM, since all the research is based on the states in the US, the extent of industrialization is likely to be constant for all states. However, as we have to apply the model towards countries that have different degrees of industrialization from the world, an **Environmental-Industrial Index (EII)** is required to enhance NIEM's comprehensiveness.

Essentially, EII is a **benefit-type index** that exhibits the **interconnection** between the **natural environment and industrial environment** within a country, we first referred to Environmental Performance Index (EPI), which is a comprehensive rating available for most countries across the world based on their environmental performance, as an **benefit-type** indicator. Additionally, we incorporated an industrial indicator by looking at different industrial sectors' **total economic values** within the country.

$$I_I = \frac{EPI_i - EPI_{min}}{EPI_{max} - EPI_{min}} \times i_i \tag{7.1}$$

Where i_i represents the **extent of industrialization** within the country.

$$i_i = \frac{ValueUSD(C)}{ValueUSD(B) + ValueUSD(C) + ValueUSD(D) + ValueUSD(E)} \tag{7.2}$$

Here, $ValueUSD(B)$, $ValueUSD(C)$, $ValueUSD(D)$, and $ValueUSD(E)$ refer to the total economic value of **mining and quarrying**; **total manufacturing**; **electricity, gas, steam, and air conditioning**; and **water supply, sewerage, and waste management**, respectively.

Arguably, $ValueUSD(C)$, as part of the industrial indicator, partially reflects how well a country can regulate its natural environment. Specifically, as the total economic value of manufacturing within a country increases, the country will have better technology and thereby regulating its natural environment more efficiently. Therefore, we put $ValueUSD(C)$ on the numerator, indicating that it is a benefit-type variable.

$$S_6 = \frac{I_{Lmax} - I_{Li}}{I_{Lmax} - I_{Lmin}} \tag{7.3}$$

Adjusted: Carbon Emission Index (CEI)

The adjusted CEI is shown as follows:

$$I_{Ci} = \frac{(\frac{|t_i - t_{comfort}|}{t_{max} - t_{min}} + 0.5) \times (c_i \times p_i + c_w \times p_w)}{(p_i + p_w) \times (\frac{\xi_i - \xi_{min}}{\xi_{max} - \xi_{min}} + 0.5) \times (\frac{A_i - A_{min}}{A_{max} - A_{min}} + 1)} \tag{7.4}$$

Where ξ_i represents the quality of water in the country; c_i represents the combination of carbon emission caused by residents and transportation.

Essentially, we added the effect of other **country-specific factors** into the formula. Notably, we did not use spatial carbon intensity per person in this formula; instead, we turned the area of the country into an insignificant factor by normalizing it.

Adjusted: Energy Consumption Index (ECI)

$$I_{Ei} = \frac{(\frac{|t_i - t_{comfort}|}{t_{max} - t_{min}} + 0.5) \times (p_i \times e_i + p_w \times e_w)}{(p_i + p_w) \times r_i \times (\frac{\xi_i - \xi_{min}}{\xi_{max} - \xi_{min}} + 0.5) \times (\frac{\upsilon_i - \upsilon_{min}}{\upsilon_{max} - \upsilon_{min}} + 1)} \tag{7.5}$$

Where υ_i represents the energy generated by wind.

Adjusted: Water Consumption Index

$$I_{Wi} = \frac{W_{ci} + W_{cw}}{(\frac{\eta_i - \eta_{min}}{\eta_{max} - \eta_{min}} + 1) \times (\frac{\xi_i - \xi_{min}}{\xi_{max} - \xi_{min}} + 1) \times W_{si} \times (\frac{p_i - p_{min}}{p_{max} - p_{min}} + 1)} \tag{7.6}$$

The availability of data enables us to enhance NIEM's reliability and accuracy.

Here, η_i represents the availability of water in the country. We have also changed the meanings of the symbols: W_{ci} refers to the total water consumed in a country; W_{si} is the total water supply in a country; and W_w is the average total water consumed in World Cup session.

Adjusted: AQI Index

$$I_{AQIi} = \frac{(AQI_i - AQI_s)^2 + 0.5}{(\frac{\iota_i - \iota_{min}}{\iota_{max} - \iota_{min}} + 1)} \tag{7.7}$$

Where, ι represents the average wind speed within the country, and AQI_s represents the standard AQI for us to perform the calculation, which we took 100 as the standard AQI value, considering the industrialization requirement.

Comparing to the CIEM, there are certain changes. Specifically, we calculated the distance from each country's AQI to the standard AQI, and w used the average wind speed within the country for calculation.

Adjusted: Landfill index

$$I_{Li} = \frac{L_w \times (\frac{3.5 \times 10^6 - p_{min}}{p_{max} - p_{min}} + 1) + L_i \times (\frac{p_i - p_{min}}{p_{max} - p_{min}} + 1) \times 32}{EPI_i \times logA_i} \tag{7.8}$$

Here, L_w represents the amount of trash and waste created per audience during a World Cup session; R_p represents the range of population within the dataset.

Normalization

Similarly, we used the same method in CIEM to normalize the 6 indices, obtaining S_i for each country:

$$S_i = \frac{I_i - I_{min}}{IQR} \tag{7.9}$$

Results of the Normalized Indices

After predicting the required data for 2029, we obtained the six indices (S_1 to S_5) for each of the seven counties that are suitable for hosting the World Cup. The normalized data are as follows:

Table 7.1: Normalized Values for Selected Countries

	Germany	Japan	Brazil	South Korea	Russia	South Africa	Qatar
S_1	0.0000	0.1350	1.1342	0.1920	1.0662	0.3858	1.2250
S_2	0.0030	0.0000	0.0691	0.0150	0.5075	1.0000	0.2402
S_3	0.2263	0.3583	0.9585	0.1508	1.0000	0.2860	0.0000
S_4	0.0000	0.1182	0.6959	0.8946	0.4324	0.4155	1.0000
S_5	0.0000	2.2061	1.5000	1.7227	0.9939	2.0789	0.7227
S_6	1.0000	0.7321	0.5796	0.5415	0.2780	0.1693	0.1000

7.2 Weighting

Analytical Hierarchy Process (AHP)

Considering the global characteristics of World Cup events, we constructed the following judgment matrix for the six indicators:

$$A = \begin{bmatrix} 1 & 3 & 5 & 7 & 9 & 4 \\ 1/3 & 1 & 3 & 5 & 7 & 2 \\ 1/5 & 1/3 & 1 & 3 & 5 & 1/2 \\ 1/7 & 1/5 & 1/3 & 1 & 3 & 1/4 \\ 1/9 & 1/7 & 1/5 & 1/3 & 1 & 1/6 \\ 1/4 & 1/2 & 2 & 4 & 6 & 1 \end{bmatrix} \tag{7.10}$$

Indicator order: S1, S2, S3, S4, S5, S6

Rationale of the judgment matrix:

Carbon emissions and Energy consumption remain the highest priority due to their long term and irreversible global climate impacts. The new EII ranks third as it reflects **comprehensive national environmental governance**. Water resources and air quality address more regional concerns, while landfill management has the most localized impact, so they were less prioritized.

Weight Calculation

The normalized weights are calculated using:

$$w_i = \frac{M_i}{\sum_i M_i}, \quad \text{where } M_i = \left(\prod_{j=1}^{6} a_{ij} \right)^{1/6} \tag{7.11}$$

Resulting weights:

$$W_s = [0.430, 0.229, 0.103, 0.057, 0.032, 0.148]^T \tag{7.12}$$

Consistency Check: $CR = \frac{CI}{RI} = 0.029$

Since $CR = 0.029 < 0.10$, the judgment matrix passes the consistency check.

Criteria Importance Through Intercriteria Correlation (CRITIC)

Our 6 new, adjusted indices from NIEM are arguably more **interconnected** with each other than the 5 indices from CIEM. For instance, the EII and the LFI share the variable EPI together. Therefore, we considered the best way to handle such interdependence to be the use of the CRITIC method. Fittingly, CRITIC decides weights based on both **variability** (measured by standard deviation) and **interdependence** (measured by Pearson Correlation Coefficient), meaning that, other than measuring interdependence, it actually 'encompasses' EWM as well.

First, we utilized **max-min scaling** to normalize all 42 **cost-type values** (6 indices value for each of the 7 different countries):

$$h_{ij} = \frac{max_j - x_{ij}}{max_j - min_j}$$

Next, to measure variability, we calculated the sample standard deviation (s) of the normalized values by the following formula:

$$s_j = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(h_{ij} - \bar{h_j})^2}$$

Subsequently, we calculated Pearson correlation coefficient (r) for the normalized sample:

$$r_{ij} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Then, we obtained the **importance of criterion (**C_j**)** and calculated the weight of each S (normalized indices of CIEM):

$$w_j = \frac{C_j}{\sum_{j=1}^{n} C_j} = \frac{s_j \cdot \left(1 - \sum_{i \neq j} |r_{ij}|\right)}{\sum_{j=1}^{n} s_j \cdot \left(1 - \sum_{i \neq j} |r_{ij}|\right)}$$

Our results for the weight are as follows:

$$W_o = [0.211, 0.119, 0.147, 0.143, 0.204, 0.175]^T$$

Strategic AHP-CRITIC Combination

We combine the **subjective AHP weights** (W_s) and **objective CRITIC weights** (W_o) in a linear fashion:

$$W_c = \alpha W_s + (1 - \alpha) W_o, \quad \alpha = 0.6 \tag{7.13}$$

Still, we set the value of α to be 0.6 (60%), meaning that W_s, the subjective AHP, is slightly more important than W_o, the objective EWM, because we deemed our research- and experience-based insights to have greater practical value.

Table 7.2: AHP vs CRITIC Weights

Indicator	**AHP Weight (W_s)**	**CRITIC Weight (W_o)**	**Combined Weight (W_c)**	**Rank**
CEI (S_1)	0.430	0.211	0.3424	1
ECI (S_2)	0.229	0.119	0.1850	2
WCI (S_3)	0.103	0.147	0.1206	4
AQI (S_4)	0.057	0.143	0.0914	6
LFI (S_5)	0.032	0.204	0.1008	5
EII (S_6)	0.148	0.175	0.1588	3
Total	**0.999**	**0.999**	**1.000**	

Verification of Combined Weights:

Total: $0.3424 + 0.1850 + 0.1206 + 0.0914 + 0.1008 + 0.1588 = 0.999 \approx 1$

7.3 Comprehensive Evaluation (TOPSIS) and Results

Again, we used TOPSIS to yield the final comprehensive evaluation; the environmental sustainability scores for potential World Cup host countries are:

Table 7.3: TOPSIS Scores for World Cup Host Country Evaluation

Country	TOPSIS Score	Rank
Germany	0.8124	1
Japan	0.6743	2
South Africa	0.5931	3
Brazil	0.5216	4
Russia	0.4872	5
South Korea	0.4638	6
Qatar	0.3925	7

Conclusion

The NIEM model identifies **Germany** as the most environmentally sustainable choice for hosting the World Cup 2030, with the highest TOPSIS score of 0.8124. This is brought about by Germany's exceptional performance across **carbon emissions, air quality, and landfill impact**.

7.4 Comparison and Conclusion

Environmental Impact Minimization Strategies

- **Opening special bus lines and shuttle buses for the competition:**

  This can directly reduce the transportation carbon emissions relative to the competition by lowing ϵ, which is the transport intensity per capita. Afterwards, it can be optimized to fit local demand and kept as a new permanent bus route, so as to permanently improve the density and efficiency of the city's public transportation network, directly reduces T_i^c and T_i^e, and prepare for future large-scale sports events at any time. This process reduces the overall ECI of the city.

- **Purchase and Invest in solar power generation equipments for the competition:**

  This action can directly improve the score of ECI in short-term, by temporarily boost r_i during the event period, making the city more suitable for hosting the sport event. After the competition, the equipments can be transferred to other places and continue to generate electricity, increasing r_i and the overall renewable score of the city permanently.

- **Implement a strict garbage classification policy and composting system at venues:**

  This action reduces the total landfill waste L generated during the event, thereby lowering the LFI in the short-term. The established policy and public awareness can then be transferred to a city-wide garbage recycling policy after the event. This permanently reduces the amount of waste that need to be disposed through unsustainable way, improves the environmental quality index e^{th}, leading to a long-term improvement in LFI score.

- **Improve stadiums' facilities with water-efficient fixtures:**

  This strategy directly reduces the water consumption multiplier γ for participants during the event, which lowers the WCI by decreasing $(80000 \times \gamma + p_i)$. After the event, the water-saving facilities, such as low-flow faucets and waterless urinals, can be promoted to other buildings across the city. This systematically reduces the city's per capita water consumption W_{ci}, increasing the $p_i \times W_{ci}$ in the WCI formula, obtaining permanent water savings and better score.

Environmental Factors Comparison

For a single-game event such as the Super Bowl (evaluated by CIEM), CEI and ECI are top priorities. Additionally, WCI also has an relatively heavy impact on the evaluation, as water consumption pressure is more prominent in smaller areas like cities.

For a multi-venue and multi-sport event like the World Cup (evaluated by NIEM), EII is added to account for the difference in the level of industrialization between countries. Moreover, weights are adjusted to account for index interdependencies,, making the weight for each index, especially CEI and ECI, relatively lower compared to that of CIEM.

8 NFL Host City Letter

Dear NFL committee,

We hope this letter finds you well.

As public awareness of mega sporting events' sustainability grows, the NFL faces increasing pressure to mitigate the environmental footprint of its well-known event—the Super Bowl. In response, we have established City Impact Evaluation Model (CIEM), a regionally specific holistic model that evaluates potential Super Bowl host cities based solely on environmental considerations. Having the goal of determining the most sustainable option for the host of Super Bowl LXIII, we have chosen Inglewood, California, after carefully obtaining required data from credible sources and performing the calculations.

Notably, CIEM considers a host city's carbon emission, energy consumption, water consumption, air quality, and landfill volume when assessing the environmental impacts. Each of the five core aspects for sustainability is expressed in an equation, denoted as an index, in which more than 14 different contributing factors that influence the index are also identified and incorporated into the equation. Therefore, CIEM effectively ensures comprehensiveness, accuracy, and practicability. Moreover, although we have not demonstrated this in our study, CIEM is highly adaptable. Its city-specific variables and indices makes it applicable to most US cities.

Inglewood, California, becomes our top choice for the NFL to consider, as it encompasses the following environmental advantages.

Inglewood yields a carbon emission index (CEI) of 0, meaning that it has the lowest carbon dioxide emission among the potential host cities tested. By hosting the next Super Bowl event in Inglewood, carbon emission can be effectively minimized, reducing global warming. Second, Inglewood's energy consumption index (ECI) is 0 as well, meaning that it has the lowest energy requirement when hosting a Super Bowl event and it actively incorporates renewable energy into use, which perfectly aligns with the NFL Green's goal of relying more on green and renewable energy sources. Third, Inglewood's water consumption index is again 0, indicating a mitigated impact by water consumption on the local environment. Fourth, Inglewood's air quality index is slightly higher than other potential hosts; however, air pollution impacts are local and temporary, which will quickly return to baseline. Last but not least, Inglewood's has a relatively low landfill index, meaning that it's environment is similar to New Orleans, which was arguably successful in mitigating its landfill build-up.

Overall, Inglewood stands out as the most environmentally responsible choice for the 2029 Super Bowl LXIII. Its exceptional performance across carbon emissions, energy use, water consumption, and landfill build-up reflects clear alignment with the NFL's sustainability goals, making it the best positioned host city to minimize the event's environmental footprint.

Yours sincerely,
Team 16442

9 Strengths and Weaknesses

Strengths

- Instead of using singly an objective or a subjective weighting method, we **strategically combined** both **Analytical Hierarchy Process (AHP)** and **Entropy Weight Method (EWM)** when creating our City Impact Evaluation Model (CIEM), which ensures fairness and reasonably reflects expert judgment, making the results balanced and robust.
- Instead of adding EWM, we melded **CRiteria Importance Through Intercriteria Correlation (CRITIC)** with AHP to adapt our National Impact Evaluation Model (NIEM), as the 6 indices in NIEM are more interconnected. This not only ensures objectivity within subjectivity but also better handles the interconnected relationships between the 6 indices.
- Our evaluation models-CIEM and NIEM-encompass **5 and 6 key environmental impact indices**, respectively, each of which is further broken down into **smaller contributing factors**, making both evaluation comprehensive and effective.
- Through sensitivity analysis, we observed that there is no significant fluctuation in our final results, proving CIEM to be robust and reliable.
- Both CIEM and NIEM are designed for US cities and countries around the world, respectively. Thus, they are arguably adaptable in most circumstances.

Weaknesses

- We applied certain **state-level data** (e.g., energy consumption per capita) to the potential host cities, which is a justified assumption. Nonetheless, this approach is only effective for **approximations**. Ideally, we should procure city level data, which we didn't found on the internet, by methods like on-site data collection to be more precise and accurate.
- The units for certain variables in both CIEM and NIEM are **non-standard**, such as Btu and degree Fahrenheit (imperial units), which may lead to misunderstanding and inconvenience when CIEM and NIEM are used by others to conduct evaluations. To make both CIEM and NIEM more accessible, we should unite every non-standard units into **SI units or other standardized units**.
- Both CIEM and NIEM assumes stable environmental policies and infrastructures, which is **overly static and realistically impossible**, hindering their practical value.

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10 AI Use Demonstration

First, we conducted several preliminary estimation with the aid of AI. For instance, we utilized ChatGPT to estimate the water consumption during Super Bowl LIX:

- **AI Model:** OpenAI ChatGPT (Nov 9, 2025 version, ChatGPT-5 Mini)
- **Query:** Based on drinking and restrooms usage, roughly estimate the total water consumption during Super Bowl LIX. Suppose there are 80000 participants (including audiences, staff members, etc). Respond with the results, in liters, ONLY.
- **Response:** Approximately 4,000,000 liters.

Second, we left certain calculation (other calculations are done by excel or python coding) for AI to perform. To illustrate, when we were substituting known or predicted data into our Carbon Emission Index (CEI).

- **AI Model:** OpenAI ChatGPT (Nov 13, 2025 version, ChatGPT-5 Mini)
- **Prompt:** Based on the formula for Ic (attached image), substitute the available data in the table (attached excel table) and calculate the Carbon Emission Index (Ic) for the following cities: Inglewood (California), Glendale (Arizona), Las Vegas (Nevada), and New Orleans (Louisiana), Charlotte (North Carolina), Chicago (Illinois), Nashville (Tennessee). After this, perform the normalization for all 7 cities (attached image, calculate S1) and present the results in a table. Respond with an LaTeX-format vector for me to copy, keeping 3 decimal places, ONLY.
- **Response:** The code generated is as follows: $S_1 = (0, 0.036, 0.178, 0.416, 0.128, 0.059, 0.453)$

Third, based on our data, we utilized AI to draw diagrams and visual aids when needed. For instance, we used AI to draw Figure 6.5.

- **AI Model:** OpenAI ChatGPT (Nov 15, 2025 version, ChatGPT-5 Mini)
- **Prompt:** Based on the attachment, which displays the sensitivity analysis performed for the number of participants or audiences during a Super Bowl event and its impact on the change in rankings and TOPSIS scores of the 7 states, create a diagram showing how city score trends vary with the number of audiences. Use different colors to represent different cities. Please provide the python code ONLY.
- **Response:**

python Copy

```python
import matplotlib.pyplot as plt
import pandas as pd

cities = ['Inglewood (CA)', 'Glendale (AZ)', 'Las Vegas (NV)', 'Charlotte (NC)',
          'Nashville (TN)', 'Chicago (IL)', 'New Orleans (LA)']

audience_scenarios = {
    '-10% (72,000)': [0.7012, 0.6815, 0.6421, 0.5753, 0.5532, 0.4528, 0.4004],
    'Base (80,000)': [0.7038, 0.6842, 0.6446, 0.5770, 0.5551, 0.4545, 0.4021],
    '+10% (88,000)': [0.7063, 0.6868, 0.6470, 0.5786, 0.5569, 0.4561, 0.4037]
}

audience_numbers = [72000, 80000, 88000]
markers = ['o', 's', '^', 'D', 'v', '<', '>', '*']
colors = ['#1f77b4', '#ff7f0e', '#2ca02c', '#d62728', '#9467bd', '#8c564b', '#e377c2

plt.figure(figsize=(14, 8))

for i, city in enumerate(cities):
    scores = [audience_scenarios[key][i] for key in audience_scenarios]
    plt.plot(audience_numbers, scores, marker=markers[i], label=city,
             linewidth=2.5, markersize=10, color=colors[i],
             markerfacecolor='white', markeredgewidth=2)
    for x, y in zip(audience_numbers, scores):
        plt.annotate(f'{y:.4f}', xy=(x, y), xytext=(5, 5 if i % 2 == 0 else -15),
                     textcoords='offset points', fontsize=9, alpha=0.8)

plt.xlabel('Number of Audiences', fontsize=14, fontweight='bold')
plt.ylabel('Environmental Impact Score', fontsize=14, fontweight='bold')
plt.title('City Score Trends with Audience Number Variations\n(Sensitivity Analysis
          fontsize=16, fontweight='bold', pad=20)
plt.xticks(audience_numbers, ['72,000\n(-10%)', '80,000\n(Base)', '88,000\n(+10%)'],
plt.yticks(fontsize=12)
plt.grid(True, alpha=0.3, linestyle='--')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left', frameon=True, fancybox=True,

df = pd.DataFrame(audience_scenarios, index=cities)
df['Absolute Change'] = df['+10% (88,000)'] - df['-10% (72,000)']
df['Relative Change (%)'] = (df['Absolute Change'] / df['Base (80,000)']) * 100
print(df.round(4))
print(f"\nAverage Absolute Change: {df['Absolute Change'].mean():.4f}")
print(f"Maximum Absolute Change: {df['Absolute Change'].max():.4f} (Inglewood)")
print(f"Minimum Absolute Change: {df['Absolute Change'].min():.4f} (Charlotte)")
print(f"Average Relative Change: {df['Relative Change (%)'].mean():.2f}%")
print(f"Maximum Relative Change: {df['Relative Change (%)'].max():.2f}% (New Orleans
print(f"Ranking Stability: Perfect across all scenarios")
```

Figure 10.1: Python Code